

Test for Single Proportion

If x is the number of individuals possessing the given attributes in a random sample of size ' n ' from a large population, then ' p ' is termed as proportion of success.

$$P = \frac{x}{N}$$

$$E(p) = P$$

$$SE(p) = \sqrt{PQ/n}$$

$$Z = \frac{p - E(p)}{\sqrt{PQ/n}} \sim N(0,1)$$

If we are having a sample from a finite population of size N , then

$$SE(p) = \sqrt{\frac{(N-n)PQ}{(N-1)n}}$$

$$\text{then } Z = \frac{p - P}{\sqrt{\frac{(N-n)PQ}{(N-1)n}}}$$

Ques: A die was thrown 9000 times. Out of this, 3220 yield a 3 or 4. Can the die be regarded as unbiased. Use 1% level of significance.

$$\text{Given } N = 9000, X = 3220,$$

$$P = \frac{X}{n} = \frac{3220}{9000} = 0.357$$

$P = E(p)$ = Population proportion of success
= Probability of getting 3 or 4 on a die

$$P = \frac{2}{6} = \frac{1}{3} = 0.33$$

$$Q = 1 - 0.33 = 0.67$$

We have to test,

$$H_0 : P = 0.33 \quad [\text{Die is unbiased}]$$

$$H_1 : P \neq 0.33 \quad [\text{Die is biased}]$$

Level of significance = 0.01

$$Z = \frac{P - P}{\sqrt{PQ/n}}$$

$$\text{ie } Z = \frac{0.357 - 0.33}{\sqrt{(0.33 \times 0.67)/9000}}$$

$$Z = 5.44$$

From table, $Z_{(1\%)} = 2.58$

Calculated value $>$ table value.

$\therefore$ We reject H_0 .

Hence we can conclude that die is biased.

Ques: In a random sample of 400 persons in a large population, 120 are females. Can it be said that males and females are in the ratio 5:3 in the population. Use 1% level of significance.

$$\text{Given } n = 400, X = 120$$

$$P = \frac{X}{n} = \frac{120}{400} = 0.3$$

$P = E(p) =$ Proportion of females in the population

$$P = \frac{3}{8} = 0.375$$

$$Q = 1 - P = 1 - 0.375 = 0.625$$

we have to test,

$H_0: P = 0.375$ [Proportion of males & females are in the ratio 5:3]

$$H_1: P \neq 0.375$$

Level of significance = 0.01.

$$Z = \frac{p - P}{\sqrt{PQ/n}} = \frac{0.3 - 0.375}{\sqrt{\frac{0.375 \times 0.625}{400}}}$$

$$= -3.09$$

$$|Z| = 3.09$$

From table, $Z = 2.58$

Calculated value $>$ table value

so H_0 is rejected.

Hence the proportion of males and females are not in the ratio 5:3.

ues: In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that majority of men in this city are smokers. Use 5% level of significance.

$$\text{Given } n=600 \quad X=325 \quad P = \frac{X}{n} = \frac{325}{600} = 0.54$$

$P = E(p)$ = Probability of a man to be a smoker.

$$P = 0.5$$

$$Q = 0.5$$

$$SE = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{(0.5)(0.5)}{600}} = 0.0204$$

we have to test,

$$H_0 : P = 0.5$$

$$H_1 : P > 0.5$$

$$Z = \frac{p - P}{\sqrt{PQ/n}} = \frac{0.54 - 0.5}{0.0204}$$

$$Z = 1.96$$

From table $Z = 1.645$

Calculated value $>$ table value. So reject H_0 .
Hence majority of men in the city are smokers.